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Open and Closed Knowledge Sourcing: Their Effect on Innovation Performance in Small and Medium Enterprises

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Open and Closed Knowledge Sourcing: Their Effect on Innovation Performance in Small and Medium Enterprises

Abstract

Purpose – Many studies have investigated the relationship between the adoption of open innovation and performance in large firms. However, limited research is available with regard to the use of open innovation in small and medium enterprises (SMEs). SMEs are important because of their contribution to innovation in almost all economies. This study extends the current literature by focusing on SMEs. Using complementarity and knowledge-based theories, this study develops three hypotheses to identify the effect of knowledge sourcing approaches for innovation on SMEs' innovation performance.

Design/methodology/approach – Surveys collected from 196 SMEs in Korea were analyzed using the supermodularity function to test the hypotheses.

Findings – Results indicate that an external knowledge-oriented approach has no significant effect, whereas an internal knowledge-oriented (i.e., closed) approach has a positive effect on innovation performance. Interestingly, this study found that open innovation has a negative effect on SMEs' innovation performance (i.e., both internal knowledge-oriented and external knowledge-oriented approaches have a substitutive relationship).

Originality/value – This study sheds new light on open innovation and knowledge management research by identifying the relationship between knowledge sourcing approaches for innovation, and innovation performance in SMEs. Practical implications highlight that open innovation could impede SMEs' innovation performance.

Keywords Open innovation, Knowledge Sourcing Approach, Complementarity, Substitutability, SMEs

Paper type Research paper

1. Introduction

Open innovation is defined as “the use of purposive inflows and outflows of knowledge to accelerate internal innovation and expand the markets for external use of innovation, respectively” (Chesbrough *et al.*, 2006). Many researchers have emphasized its strategic importance as a critical approach for innovation, and have suggested that a positive relationship exists between open innovation and a firm’s performance (Christensen *et al.*, 2005). In addition, many firms in a variety of industries, such as the high-tech, service, and chemical industries, have adopted open innovation to improve their performance (Chesbrough, 2011).

The basic assumption of open innovation is that by using external knowledge in conjunction with internal knowledge, the performance of a firm can improve in a rapidly changing environment (Chesbrough, 2003). Leiponen and Helfat (2010) suggested that firms can improve their odds of achieving successful innovations by accessing to broad knowledge sources, including external ones. Rass *et al.* (2013) proposed that open innovation instruments positively affect a firm’s performance through the firm’s social relations and networks. Procter & Gamble estimated that they can save more than 25% of their printing and copying costs by introducing open innovation (Chesbrough, 2011). For example, Curana, a small bicycle accessory manufacturer in Belgium, has become a trendsetter in the bicycle industry, and most bicycle manufacturers wish to combine their products with Curana’s products. Open innovation enabled Curana to develop new and innovative products, such as mudguard combining aluminum and plastic, and made the firm a worldwide market leader (Vanhaverbeke *et al.*, 2012).

However, little research has verified if open innovation indicates a better performance for a firm compared with the use of closed innovation despite the theoretical entrenchment of the idea that firms can use internal and external knowledge sources together. Some studies reveal that a firm’s performance can increase by using closed innovation and focusing on the internal R&D investment because doing so enables the firm to have an in-depth understanding of the strengths and weaknesses of its own knowledge and capabilities (Bogner and Bansal, 2007). Other studies pointed out that open innovation, which focuses on combining internal and external knowledge sources, can enhance a

firm's performance by allowing the firm to have a broader range of knowledge, and enabling the firm to have a different perspective on its critical issues (De Clercq and Dimov, 2008). To verify the accuracy of the previous studies' results, many efforts have been made to investigate the conditions where the closed and open innovation approaches are applicable (Garrig *et al.*, 2013; Makri *et al.*, 2010). Almirall and Casadesus-Masanell (2010) argued that the open innovation approach restricts a firm's ability to establish technological capabilities when suppliers and/or complementors have different goals.

Although previous studies on open innovation improved the understanding of the open innovation phenomenon, these studies were performed specifically for managers in large firms. Only a few studies focused on open innovation in small and medium enterprises (SMEs). Open innovation in SMEs is significant for the following reasons. First, SMEs are important because of their contribution to innovation in almost all economies in the world. According to the Organization for Economic Cooperation and Development (OECD), SMEs represent more than 95% of enterprises and account for 60%–70% of employment in most OECD countries (OECD, 2005). Therefore, encouraging innovation in SMEs is essential for policy makers to promote economic growth (Lee *et al.*, 2010).

Second, SMEs often lack human capital, financial resources, and knowledge to develop and commercialize new products or services internally. To overcome their inherent limitations, SMEs opt to collaborate with other organizations. SMEs have thus engaged in various open innovation practices and continue to adopt such practices (Dufour and Son, 2015). Furthermore, open innovation in SMEs has become more important than before (Van de Vrande *et al.*, 2009). Open innovation is a critical issue for SMEs to reduce innovation risk and cost and increase operation flexibility in today's rapidly changing and knowledge-intensive world with shortened product and service life cycles (Lee *et al.*, 2010).

Third, major differences exist between large enterprises and SMEs (Gopalakrishnan and Bierly, 2006). Therefore, different open innovation approaches need to be adopted depending on firm size (Vossen, 1998). Significant differences exist between large firms and SMEs in terms of the resources that constrain a firm's ability to access, acquire, and use knowledge (Wong, 2005). Therefore, simply

viewing SMEs' open innovation approaches as scaled-down versions of those used in larger firms is incorrect (Pillania, 2008).

To explore the research gaps in previous studies, this study examines the effect of a firm's size on its innovation performance from the perspective of SMEs by focusing on knowledge sourcing. Open innovation emphasizes the importance of using external and internal knowledge for a firm's innovation (Torres *et al.*, 2015). Therefore, this study focuses on how SMEs' innovation performance varies depending on its knowledge sourcing approaches. The term "knowledge sourcing approach" is defined as the extent to which an individual or organization intentionally accesses, acquires, and reuses knowledge from others, such as expertise, experiences, insights, and opinions (Gray and Meister, 2004). For this purpose, this study first proposes three different approaches of innovation: the internal knowledge-oriented approach, the external knowledge-oriented approach, and the open innovation approach (i.e., combining internal knowledge-oriented and external knowledge-oriented approach). This study then theoretically explores the effect of the open innovation approach on the firm's innovation performance by using complementarity theory (Milgrom and Roberts, 1995). The complementarity approach helps to conceptualize the notion of synergy, which states that the valuation of a combination of activities exceeds the sum of valuations for standalone activities (Massini and Pettigrew, 2003).

2. Literature Review

Innovation is considered a source of sustained competitive advantage for firms in a dynamically changing environment (Crossan and Apaydin, 2010). Therefore, the innovation process has duly prompted and inspired many studies in the business field. Researchers have devoted increased attention to innovation and its different facets, especially in the last decade, because of the recognized value of innovation. As a result, extant literature on innovation is vast and covers a broad range of perspectives for firms within a diverse assortment of industries. Past research on innovation can be divided into three categories based on the knowledge sourcing origin. These three are internal knowledge oriented, external knowledge oriented, and open innovation (i.e., combining both internal

and external knowledge oriented) approaches. Appendix A shows a comparison of these different innovation approaches and key extant studies.

Previous studies under the first category focus on using internal knowledge for a firm's innovation. The internal knowledge-oriented approach (i.e., closed or traditional approach) emphasizes the importance of using internal knowledge in improving a firm's innovation performance (Kessler *et al.*, 2000). Knowledge developed and shared within a firm helps in the development of its core competencies and capabilities. The knowledge developed and shared in the firm allows the firm to reap more profits from innovation efforts because the internal knowledge is likely to be unique and specific to the firm (Choi and Lee, 2012). Thus, copying or imitating any innovation from the firm is difficult for competitors; thus, the firm's existing knowledge and the development of new opportunities for business are protected. Bogner and Bansal (2007) observed that when a firm develops new knowledge by building on its own internal knowledge, the firm's superior value is attained.

Studies under the second category highlight the significance of using external knowledge for a firm's innovation performance (Grevesen and Damanpour, 2007). External knowledge helps firms to develop a broader knowledge base and to remain up to date with the latest emerging technologies (Bierly and Chakrabarti, 1996; Grant, 1996). In addition, external sourcing of knowledge helps firms to reduce the costs of its development and increases the firm's speed of implementing its newly acquired knowledge (Ai and Wu, 2016; Gopalakrishnan and Bierly, 2001). Therefore, the ability of a firm to absorb external knowledge is crucial for the firm's innovation capabilities (Cohen and Levinthal, 1990). Bierly and Daly (2007) empirically tested the relationship between a firm's external knowledge and innovation. They concluded that a firm's use of external knowledge positively affected the firm's innovation speed. In addition, they concluded that SMEs used different knowledge sources from larger firms.

Studies under the third category focus on both internal and external knowledge to increase a firms' innovation performance (Hoang and Rothaermel, 2010). Adopting both knowledge sources helps firms to acquire new knowledge effectively through external means (Katila and Ahuja, 2002), while allowing firms to build rare and valuable capabilities to improve their innovative performance (Wu

and Shanley, 2009). Zahra and Nielsen (2002) analyzed the effect of the use of internal and external knowledge sources on multiple dimensions of technology commercialization. They found that the integration of a firm's internal and external knowledge sources strengthens the contributions of the firm's manufacturing sources to its innovation performance.

Several interesting inferences can be made by combining facts from previous studies. First, although the extant literature on knowledge sourcing for innovation is vast, the empirical studies mostly focused on large firms. Few previous empirical studies focused on the knowledge sourcing required for SMEs' innovation (Lee *et al.*, 2010). Second, the relationship between open innovation and firm performance is indefinite. Although many authors have theoretically suggested that the open innovation approach can increase a firm's innovation performance, empirical studies revealed inconsistent results. While some studies have suggested that a firm's enhanced performance (i.e., complementarity) comes from adopting the open innovation approach (Cassiman and Veugelers, 2006), others have argued that the open innovation approach can impede a firm's performance (Nevo *et al.*, 2007). Finally, few studies considered the performance implications of adopting the open innovation approach on SMEs. The results of the studies that focus on large firms cannot be used to make similar judgments with SMEs. Thus, firms need to understand the performance gains they can obtain when adopting the open innovation approach before they decide to implement it within their firms.

3. Theoretical Background and Hypothesis Development

3.1. Knowledge-based theory

Originating from previous strategic management literature, knowledge-based theory (KBT) is built upon the concept of resource-based theory (RBT). According to RBT, firms can achieve a sustained competitive advantage over their competitors provided that they have valuable, rare, inimitable, and non-substitutable resources (Barney, 1991). RBT is widely accepted, acknowledged, and used by scholars around the world to explain the behaviors of various firms. However, it is often criticized for viewing knowledge as a generic resource as opposed to being a resource with special attributes (Grant, 1996). In addition, the theory does not reveal the source of the rare and valuable resources that

provide the firm's competitive advantage (Bogner and Bansal, 2007). KBT attempts to answer these criticisms.

KBT states that the services provided by a firm's tangible resources depend on how they are collectively shared and applied, which is in turn a function of the firm's know-how, i.e., knowledge (Alavi and Leidner, 2001). KBT of the firm considers knowledge the most strategic and significant resource of the firm (Grant, 1996). KBT's proponents argue that because knowledge-based resources are usually difficult to imitate and are socially complex, these diverse knowledge bases and capabilities within firms are the major determinants of the firm's sustained competitive advantage and its superior performance (Alavi and Leidner, 2001). Firms that can develop, manage, and transfer this valuable, rare, inimitable, and irreplaceable knowledge are more likely to gain a sustained competitive advantage in the long run (Saarenketo *et al.*, 2004). This knowledge is embedded and carried through multiple entities, including the firm's organizational culture and identity, policies, routines, documents, systems, and employees (Alavi and Leidner, 2001).

A critical contribution of KBT is the identification of two different knowledge sources for innovation: internal oriented and external oriented (Menon and Pfeffer, 2003). KBT considers internal-oriented and external-oriented sourcing as distinct choices. Thus, the forces that motivate a firm to use internal-oriented knowledge sourcing may not be the same as the forces that prevent it from using external-oriented knowledge sourcing (Parmigiani, 2007). A substantial number of studies have discussed the relationships between knowledge sourcing and innovation performance based on the internal-oriented and the external-oriented sourcing aspects (Wang *et al.*, 2016). Therefore, selecting among the internal-oriented, external-oriented, or the simultaneous sourcing approach (i.e., open innovation) should be one of the firm's primary decisions (Parmigiani and Mitchell, 2009; Zhou and Li, 2012).

3.2. Complementarity and substitutability theory

The notion of complementarity is commonly attributed to Edgeworth (1881). However, complementarity was formalized by the notion of supermodularity was given by Topkis (1978) and

later expanded by Milgrom and Roberts (1995). The property of supermodularity in its ordinal form is narrowly associated with the single crossing property, which states that an increase in the value of the objective function due to an increase in one of the variables is preserved even after the rise of any other variable (Mohnen and Roller, 2005). In its differential form, the single crossing property articulates that the marginal gain with respect to one variable is non-decreasing in other variables. This notion is popularly known as complementarity. Milgrom and Roberts (1995) formally defined the relationship between supermodularity and complementarity: given a real-valued function f on a lattice X , f is supermodular and its arguments are (Edgeworth) complements, if and only if, for any x and y in X , $f(x) - f(x \wedge y) \leq f(x \vee y) - f(y)$. The mathematical definition of complementarity can be described as follows: Consider a utility function of n variables, $f(x_1, \dots, x_n)$ with positive marginal utilities given by the first partial derivatives $\partial f() / \partial x_i$. Then, variables i and j are complements if an increase in the amount of one variable increases the marginal utility of the other variables. Mathematically, this concept may be represented as

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad \partial^2 f / \partial x_i \partial x_j \geq 0, \text{ for all } x \in \mathbb{R}^k, i \neq j. \quad (1)$$

The concept of substitutability is the exact opposite of the concept of complementarity.

3.3. Developing the hypothesis

This study proposes three hypotheses on the relationship among the innovation approaches and innovation performance based on KBT, and complementarity theory. Figure 1 summarizes the hypotheses.

According to KBT, knowledge is an organization's most essential resource and the only resource that cannot be easily emulated by rivals (Grant, 1996). Therefore, knowledge is an indispensable source of a firm's uniqueness or its competitive advantage (Alavi and Leidner, 2001). As SMEs cannot directly compete with larger firms on economies of scale, they pursue niche strategies by serving the narrow and smaller market segments that the larger players do not reach or fulfill (Ebben and Johnson, 2005). This objective is achieved by SMEs by focusing on market segments that require

customized products and higher levels of customer service that may be difficult for larger (generalist) firms to offer (Ebben and Johnson, 2005). Hence, these firms develop specific heterogeneous knowledge resources and capabilities that are difficult to imitate, thereby enabling the firms to achieve a sustained competitive advantage and superior innovation performance. This concept is formalized in the following hypothesis:

H1: *Internal knowledge sourcing is positively associated with SMEs' innovation performance.*

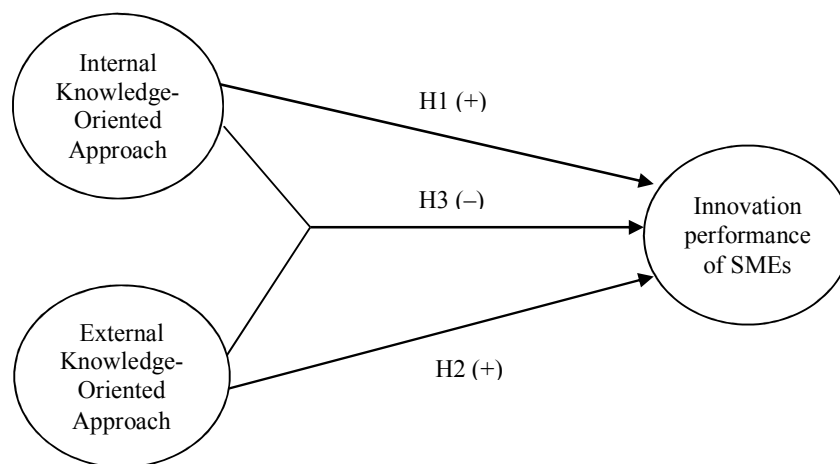


Figure. 1. Research Model

According to KBT, external knowledge enables firms to acquire fresh ideas to complement their knowledge base, thereby enhancing the firm's performance (Bierly and Daly, 2007). SMEs are typically firms that have restricted financial and human resources for innovation activities (Lim and Klobas, 2000). These firms do not have the assets to continually develop knowledge internally. In addition, developing knowledge internally can sometimes be time consuming, which may mean a lesser competitive advantage to the firm (Haas and Hansen, 2005). External knowledge is readily available to many competitors and can therefore be acquired easily at a minimal cost (Tiwana and Bush, 2007). This kind of knowledge helps firms to increase their knowledge base and to be more flexible (Zahra and Nielsen, 2002). Externally obtained knowledge helps a firm to understand how and what its competitors are doing to remain competitive in the market. Thus, this type of knowledge helps SMEs to bridge the gap between themselves and their competitors who have some competitive advantages over them. In addition, by acquiring external knowledge, SMEs can bridge their

knowledge gap at a minimal cost and time. An external knowledge-oriented approach provides firms with economies of scale in acquiring knowledge, thereby increasing the innovation performance of these firms. This concept leads to the following hypothesis:

H2: *External knowledge sourcing is positively associated with SMEs' innovation performance.*

Prior studies have highlighted the fact that adopting both internal knowledge-oriented and external knowledge-oriented approaches is complementary in large firms. The adoption of both approaches aid in increasing the large firm's innovation performance to a greater extent when carried out together as opposed to pursuing them exclusively (e.g. Cassiman and Veugelers, 2006). However, this situation might not necessarily be true for SMEs. Smaller companies may have fewer opportunities to access external knowledge because of fewer interactions with the external environment and fewer opportunities to apply this knowledge internally because of the smaller portfolio of their activities (Almeida *et al.*, 2003). SMEs are bound by their resources, capital constraints, lack of management expertise, and absorptive capacity (Forbes and Milliken, 1999). Resources and capacities are therefore important factors to be considered when implementing knowledge sourcing approaches in SMEs. Given the lack of resources and capacities, SMEs encounter considerable difficulty in integrating the two sourcing approaches seamlessly because implementing the internal knowledge sourcing is often as difficult as implementing the external knowledge sourcing, and vice versa. Although the integration of the different specialized knowledge sources is critical, high levels of knowledge sourcing in many areas are expensive, thus making it often unrealistic for SMEs (Kessler *et al.*, 2000). Therefore, SMEs that attempt to adopt an open innovation approach (i.e., combining internal knowledge- with external knowledge-oriented approaches) tend to be at a disadvantage, thereby reducing firm performance. The following hypothesis is therefore proposed:

H3: *The open innovation approach (i.e., pursuing both internal and external knowledge sourcing simultaneously) is negatively or less positively associated with SMEs' innovation performance.*

4. Research Methodology

4.1. Sampling and data collection

As a sampling frame, 926 SMEs were identified from the Annual Corporation Report in Korea. Survey questionnaires were mailed to the owner or senior manager of each firm depending on who was the most knowledgeable about the company's innovation approach and performance¹. Out of the 926 mailed surveys, 213 respondents completed the questionnaire. After eliminating the surveys with missing values, the final count was 196 firms, which provided an approximate response rate of 23%. This response rate conforms well to the typical rate for firm-level studies of approximately 20% (Paxson *et al.*, 1995). The majority (42.05%) of the firms had a total number of employees in the range of 50 to 99. Table 1 summarizes the firms' characteristics in the sample.

Table 1. Sample characteristics

(a) Industry			
Industry Type	Number of Firms	Percent (%)	Cumulative Percent (%)
Manufacturing	60	30.6	30.6
Finance	6	3.1	33.7
Service	130	66.3	100
Total	196	100	

(b) Size			(c) Age		
Firm Size	Frequency	Percent (%)	Firm Age	Frequency	Percent (%)
Less than 50	35	23.86	Less than 10	134	68.4
50 to below 100	116	42.05	10 to below 30	26	13.3
100 to below 300	36	27.84	30 to below 50	15	7.7
300 to below 500	9	6.25	More than 50	21	10.7
Total	196	100.00	Total	196	100.00

4.2. Reliability and validity for measures

Research constructs were developed based on related studies and refined through a pilot test. The operational definitions of the instruments are described in Table 2. The knowledge sourcing approaches pursued in a firm were measured based on previous studies (Zahra and Nielsen, 2002).

¹ Partial correlation was applied to determine the presence of common method bias (Podsakoff and Organ, 1986). The hypotheses were test to determine if relationships exist between the variables of innovation approaches and innovation performance after the common method factor has been statistically controlled. Given that no significant changes in regression coefficients and R² were observed, common method bias is unlikely to be a major concern.

Innovation performance was assessed using constructs developed by previous studies (Alegre *et al.*, 2006; Urgal *et al.*, 2013).

Table 2. Operational definitions and related literature

Variables	Operational Definition	Key Studies
External-oriented	The degree to which a firm seeks and uses knowledge from outside the firm for innovation. A large portion of new knowledge for innovation in my company comes... 1. ... from customers 2. ... from suppliers 3. ... from collaboration and alliance with external institutions or organizations. 4. ... from external consultants	(Kyriakopoulos and Ruyter, 2004; Nevo <i>et al.</i> , 2007; Prabhu <i>et al.</i> , 2005; Zahra and Nielsen, 2002)
Internal-oriented	The degree to which a company depends on its internal knowledge to develop new knowledge (e.g., products or services). To create and develop new knowledge for new innovations, my company frequently uses knowledge from... 1. ... colleagues 2. ... internal documents 3. ... existing products/service 4. ... internal forums/(electronic) communities	(Bierly and Chakrabarti, 1996; Kyriakopoulos and Ruyter, 2004; Teigland and Wasko, 2003)
Innovation performance	The degree of more flexible product/service provision; having faster, newer, product/service provision; having a higher-quality product/service provision; having a larger market share in comparison with major competitors. Compared with key competitors, my company 1. ... provides a wider range of products/services. 2. ... provides a new product/service more quickly. 3. ... provides a higher quality of product/service. 4. ... has a greater market share.	(Alegre <i>et al.</i> , 2006; Urgal <i>et al.</i> , 2013)
Age	The number of years a firm has existed	(Zahra and Nielsen, 2002)
Industry type	0 if the firm belongs to the service industry	(Cassiman and Veugelers, 2006; Santoro and Bierly, 2006)
	1 if the firm belongs to the manufacturing industry	
	2 if the firm belongs to the finance industry	

Controlling factors that may influence the relationship between the knowledge sourcing approach used for innovation and the innovation performance is important. First, the firm age, which is measured by the number of years a firm has existed, was controlled because firms that have been longer in the market typically have more access to external sources and greater internal sourcing (Gopalakrishnan and Bierly, 2006). Second, the study controlled for industry type. Different industry environments can have varying effects on firm performance (Santoro and Bierly, 2006), thereby affecting the firms' extent of internal use versus their external knowledge sources (Teece, 1986).

Content validity of the instrument was established through the adoption of an instrument that had already been used and validated by other researchers. A discussion was also conducted with several

professionals in the innovation and knowledge management area to confirm the content validity. All variables were measured by multi-item constructs by using a seven-point Likert scale that ranges from “very low” to “very high.” Cronbach’s alpha was used to assess the reliability of the instruments. A higher cutoff value of 0.7 was used because this had been adopted previously (Nunnally, 1978). For convergent validity, items that have item-to-total correlation scores lower than 0.5 were dropped from further analysis. Factor analysis was used to check the discriminant validity (Kerlinger, 1986). Factor analysis with varimax was adopted to check the unidimensionality among the firm performance items. Items with factor loading values lower than 0.5 were deleted. Table 3 presents the results of the reliability and validity tests, and Table 4 shows the factor analysis results.

Table 3. Statistics for reliability and validity tests

Measure	Acronym	items	Mean	S.D.	Reliability (Cronbach alpha)	Convergent Validity (Correlation of item with total score-item)	Discriminant Validity (Factor loading on single factors)
Knowledge sourcing approach	KSS	2					
Internal knowledge-oriented	IKO	4	0.371	1.463	0.929	0.799; 0.818 0.843; 0.881	0.944; 0.935 0.934; 0.916
External knowledge-oriented	EKO	4	0.342	1.596	0.950	0.899; 0.883 0.881; 0.852	0.886; 0.898 0.914; 0.936
Performance							
Innovation performance	IP	4	3.867	1.023	0.916	0.809; 0.867 0.848; 0.710	0.895; 0.930 0.919; 0.826
Control variables							
Age	AGE	1	16.79	15.17	NA	NA	NA
Industry types	INT	1	0.370	0.543			

Table 4. Factor analysis

	Factor 1	Factor 2	Factor 3
IKO 1	.373	.814	.292
IKO 2	.260	.850	.310
IKO 3	.355	.843	.213
IKO 4	.390	.801	.199
EKO 1	.792	.380	.104
EKO 2	.851	.274	.110
EKO 3	.864	.230	.235
EKO 4	.848	.344	.203
IP 1	.147	.323	.821
IP 2	.136	.227	.889
IP 3	.150	.180	.890
IP 4	.157	.118	.816
Variance explained (%)	58.31	16.80	8.47
Cumulative variance (%)	58.31	75.11	83.58

Note: IOK: Internal knowledge-oriented, EKO: External knowledge-oriented, IP Innovation performance

4.3. Testing Approach: Complementarity and Substitutability

As suggested by Athey and Stern (1998), testing for complementarity can be achieved by using two different econometric approaches. The first approach, which is called the correlation approach, tests conditional correlations on the basis of the residuals of reduced-form regressions of the organizational practices on observable exogenous variables (Mohnen and Roller, 2005). Although many prior studies have used this approach to statistically analyze complementarity mainly because of its simplicity (Arora, 1996), it cannot serve as a definitive test. Estimated correlations between residual terms may be the result of common omitted exogenous variables or measurement errors in the correlation approach. A reliable correlation exists among the practices, although it does not guarantee that the combinations of the practices lead to an enhanced output (Lokshin *et al.*, 2004). More importantly, this approach cannot disentangle interactions between more than two variables (Arora, 1996; Mohnen and Roller, 2005).

The second one is the productivity approach, which regresses a measure of productivity on a set of regressors, including the interaction effect between the different practices, and the coefficients of the interactions can be viewed as estimates of the complementarity parameters (Laursen and Mahnke, 2001). This approach tests complementarity by investigating whether the production function is supermodular. Unlike the first approach, the second approach can provide a statistical resolution for complementarity. Thus, it has been widely used in recent empirical work (Cassiman and Veugelers, 2006; Mohnen and Roller, 2005). The productivity approach represents the appropriate complementary approach for this study to analyze the fit of the knowledge sourcing approach for innovation.

This study directly estimates the contribution of a combination of two knowledge sourcing approaches for innovation to the relevant outcome measure by using supermodularity and submodularity based on the productivity approach (Arora, 1996). To formalize the hypotheses, a general production function for the firm is specified as follows: the firm maximizes a performance measure, $f(x)$, with respect to the vector of two sourcing strategies, $x = (\text{internal knowledge-oriented, external knowledge-oriented})$.

As the knowledge sourcing approaches for innovation are continuous variables, the interaction terms can be used in the regression framework to test for the sign of the interaction parameters (Mohnen and Roller, 2005). When the practices are measured by continuous values, the following definition of complementarity holds:

$$f(x_1, x_2) = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \alpha_{12} x_1 x_2 \quad (2)$$

$f()$ is the innovation performance for firm i , measured as the subjective self-reported items, and is a widely used measure of innovation performance (Alegre et al., 2006; Urgal et al., 2013). Thus, $\alpha_{12} > 0$ (i.e., the coefficient of partial derivatives $\partial^2 f / \partial x_1 \partial x_2$ is positive), implies that complementarity exists for internal knowledge-oriented and external knowledge-oriented approaches, whereas $\alpha_{12} < 0$ means substitutability for the two approaches.

5. Analysis and Discussion

5.1. Knowledge sourcing approaches and innovation performance

The data were analyzed in three steps through hierarchical regression. Table 5 provides a summary of the analysis results. In the first step, only control variables (industry and age) were analyzed in the regression model. Results show that age and industry type did not affect firm innovation performance.

In the second step, both the control variables and internal knowledge-oriented and external knowledge-oriented approaches were included in the regression model to analyze the main effect on firm innovation performance. Among the approaches, the internal knowledge-oriented approach significantly affected firm performance ($\beta=0.293$, $p<0.01$). The results thus provided strong support for hypothesis 1, which stated that the internal knowledge-oriented sourcing approach positively affects the innovation performance of SMEs. However, the external knowledge-oriented sourcing approach did not significantly affect firm innovation performance ($\beta=0.051$, $p>0.1$). Therefore, hypothesis 2 was rejected. In the last step, the interaction term was added to the existing model. Given that main effects tend to be highly correlated with the interaction terms, partial coefficient distortion for the main effects occurs when the main effects and interaction terms are analyzed simultaneously.

Using the uncorrected interaction term expectedly yielded a high variance inflation factor (VIF) of 31.412. The problem was corrected by using the residual centering technique (Lance, 1988)². Residual centering can be conducted in two stages. First, the interaction term is regressed to its respective first-order effects (i.e., internal knowledge-oriented and external knowledge-oriented approaches). Second, the residuals of the regression are used as the interaction terms instead of the original interaction terms. Analysis results obtained by using residual centering showed that the interaction term between internal knowledge-oriented and external knowledge-oriented approach was negative and significant ($\beta=-0.127$, $p<0.05$), which supports hypothesis 3. This finding implies that the open innovation approach might not be beneficial for SMEs. Table 5 showed a positive effect of the internal knowledge-oriented approach on the innovation approach in step 3 ($\beta=0.291$, $p<0.01$). However in step 3, the main effects cannot be interpreted in the presence of the interaction terms because the main effects are conditionalized (Jaccard and Turrisi, 2003).

Table 5. Analysis results

Variables	Step 1	Step 2	Step 3
Constant	3.846 [‡]	2.584 [‡]	1.723 [‡]
Step 1			
Age	0.004	0.001	0.001
Industry type	-0.122	0.000	-0.034
Step 2			
IKO (Internal knowledge-oriented)		0.293 [‡]	0.291 [‡]
EKO (External knowledge-oriented)		0.051	0.050
Step 3			
Open Innovation (IKO* EKO)			-0.127 [†]
F-Value	0.879	18.858 [‡]	16.444 [‡]
Adjusted R ²	0.000	0.269	0.285
Change in R ²		0.269	0.016
Partial F (Change in R ²)		36.512 [‡]	5.144 [†]

Note: *, †: $p<0.1$, ‡: $p<0.05$, ‡: $p<0.01$

5.2. Discussion

The test results support hypothesis 1, that is, adopting an internal knowledge-oriented approach for innovation obtains a higher innovation performance. Some studies suggest that an internal knowledge-oriented approach can cause a firm to rely on a single or a few knowledge sources to develop new

² Hypothesis 3 was also tested by using the uncorrected interaction term. The result showed that the open innovation approach (i.e., interaction term between internal and external knowledge-oriented) is negative and significant ($\beta=-0.072$, $p<0.05$), which also supports hypothesis 3.

knowledge. This approach exhibits a dysfunctional downside that inhibits a firm's innovative progress (Leonard-Barton, 1992). However, the results of this study showed the positive effect of using an internal knowledge-oriented approach on innovation performance. Focusing on internal knowledge allows firms to have an in-depth understanding of the strengths and weaknesses of their unique knowledge and skills, thereby giving them more opportunities to succeed in innovation-related activities (Bogner and Bansal, 2007; Kessler et al., 2000). Internal knowledge can also be easily accessed at a minimal cost to be used by the firm's organizational members. By contrast, competitors experience difficulty in accessing and using the firm's internal knowledge because of legal and technical barriers (Menon and Pfeffer, 2003). Therefore, the firms that improve their products and services systematically by means of internal knowledge can outperform the firms that do not use this approach (Nagarajan and Mitchell, 1998).

Contrary to our expectation, hypothesis 2 is not supported; that is, an external knowledge-oriented approach for innovation in SMEs does not show any significant positive effect on innovation performance. This intriguing result can be explained with "buy-in" syndrome, in which a firm's attitude toward external knowledge sources is more favorable than the ideal economic attitude (Lichtenthaler and Ernst, 2006). Researchers have argued that externally focused knowledge sourcing functions as a double-edged sword. Broadening a knowledge base based on external knowledge creates core competences, but at the same time, excessive application of an external knowledge-oriented approach leads to behaviors that are blind to internal knowledge. The buy-in syndrome can reduce the ability of firms to differentiate themselves from their competitors (Lichtenthaler and Ernst, 2006), and thus results in the loss of their competitive advantage (Gopalakrishnan and Bierly, 2006). Belderbos *et al.* (2006) insisted that SMEs could face diseconomies in their excessive pursuit of external knowledge because of the high costs and complexity of coordination and communication efforts to acquire external knowledge.

Our results support hypothesis 3. Some studies state that SMEs are better equipped to integrate internal knowledge-oriented and external knowledge-oriented approaches for innovation as they are more amenable to external new knowledge (Gopalakrishnan and Bierly, 2006; Van de Vrande *et al.*, 2009). As a result, the "not-invented-here" (NIH) syndrome (Katz and Allen, 1982) or core rigidities

(Leonard-Barton, 1992) are avoided. However, our results show a substitutable relationship between internal knowledge-oriented and external knowledge-oriented approaches in SMEs, as expected. SMEs have less assortment of knowledge assets and even fewer opportunities to access and acquire knowledge across the boundaries of their firms. Thus, the firms also have fewer chances of applying external knowledge internally. Moreover, a lack of organizational and managerial resources prevent SMEs from combining existing internal knowledge with new, internally created knowledge or externally acquired knowledge to improve their innovation performance. This combination requires significant resources (Almeida et al., 2003). Therefore, in SMEs, open innovation does not increase innovation performance.

5.3. Implications

The results have several implications for researchers. This study has attempted to focus on knowledge sourcing approaches for innovation, specifically for SMEs, which have often been neglected in prior studies (Lee *et al.*, 2010). Despite the vast extant literature on open innovation in large firms (Torres *et al.*, 2015), studies that identify open innovation in SMEs are lacking. Thus, the findings of this study are contrary to the conventional wisdom that states that the adoption of open innovation increases a firm's innovation performance. In fact, this study identifies a negative effect when open innovation is adopted in SMEs. The results of this study suggest that pursuing both internal knowledge-oriented and external knowledge-oriented approaches in SMEs corresponds to a substitutable effect on a firm's innovation performance rather than a complementary effect. These findings are significant for SMEs that want to innovate, as well as for knowledge management literature as this study draws a clear differentiation among the innovation performance of firms of different sizes that adopt similar innovation approaches. Hence, this study indicates a need for organizations of different sizes to customize and adopt appropriate innovation approaches for their particular firm size. Second, this study extends existing knowledge by examining the complementarity of the two types of knowledge sourcing approaches for innovation. Most prior studies investigated only the independent effects of these approaches on a firm's innovation performance without considering their combined effects. While a few studies, such as Zahra and Nielsen (2002)'s,

considered the different knowledge sourcing approaches for innovation together, they only investigated the independent effects of these approaches on innovation performance particularly in the field of technology commercialization. This study analyzes two different knowledge sourcing approaches for innovation simultaneously and provides empirical evidence on the substitutable relationships among them. Furthermore, this study tests the complementarity and substitutability of the knowledge sourcing approaches on the basis of the productivity approach. The productivity approach has been considered the definitive test method for complementarity and substitutability.

Our findings also provide valuable implication for managers in SMEs. First, our findings prove that adopting a combination of internal and external knowledge sourcing approaches for innovation is not a good initiative for SMEs. Further studies should be conducted in this field because our results clearly indicate that managers of SMEs should focus on an internal knowledge-oriented approach for their firm's innovation. In addition, pursuing an internal knowledge-oriented approach (i.e., a closed approach) requires a relatively lower budget than that needed to pursue open innovation (conducting internal and external strategies simultaneously). Concentrating on the use of a single approach helps firms remain focused on their innovation activities (Gopalakrishnan and Bierly, 2006). Pursuing a single knowledge-oriented approach will also help SMEs customize their preferred innovation approach with the organization goals easily and proficiently, thus increasing the firm's capability to maximize the benefit from newly developed expertise, as well as its pre-existing capabilities. Second, our findings suggest that adopting an external knowledge-oriented approach for innovation in SMEs does not exert any significant effect on innovation performance. SME managers attempt to assess the value and potential deficiencies of external knowledge as well as difficulties of applying this knowledge inside the firm (Lichtenthaler and Ernst, 2006). Managers should carefully exert coordination and communication efforts to acquire and evaluate external knowledge by implementing appropriate incentives for internal knowledge sourcing and by reducing disadvantageous dependence on external knowledge sources.

6. Conclusion

Firms should carefully choose their knowledge sourcing approaches for innovation to establish and maintain a sustained competitive advantage over their competitors. The results of this study suggest that firms should adopt a particular knowledge sourcing approach depending on its size. Particularly, the results of this study suggest that open innovation impedes SMEs' innovation performance, whereas closed innovation helps to enhance SMEs' innovation performance. Thus, firms must exercise due caution when adopting open innovation.

Several limitations of this study should be noted, as some of them offer opportunities for future research. First, although the self-reported performance measure of this study allows a more comprehensive analysis that is otherwise possible, the results may be dependent on the type of innovation performance measure used. A future study could examine the performance of SMEs' innovation performance by using a more quantitative approach that would provide for more factual results. Second, this study uses cross-sectional data. Future researchers are encouraged to use longitudinal data to fully understand the effect of knowledge sourcing approaches for innovation on SMEs' innovation performance. Therefore, another method to advance this research is to develop and test similar models using longitudinal data. Third, the knowledge sourcing approaches and the innovation performance data used in this study were gathered from a single key informant for each of the surveyed firms. Although information from a single executive who is responsible for the firm's innovation and knowledge management should provide a high level of confidence in the quality of the collected information, selection bias could still exist. A better picture would emerge if multiple respondents are adopted. Finally, the gathered data and the results of this study are limited to Korean firms. Thus, the application and generalization of the results of this study to other countries may be questionable. The replication of this study in other countries is vital to ensure the validity and generalizability of this study's findings.

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Appendix A. Comparison of Different Innovation Approaches with Key Extant Studies

Innovation approach	Main arguments	Selected studies		
		Researchers	Firm size	Industry
Internal knowledge-oriented	Internal knowledge is likely to be unique and specific to the firm, thus internal knowledge-oriented approach helps the development of firm's core competences and capabilities;	Bogner and Bansal (2007)	Large firms	Pharmaceuticals, semiconductors, forest products, oil and gas, and automotive
		Leiponen (2006)	Large firms and SMEs	Service
	Internal knowledge makes it difficult for competitors to copy or imitate specific knowledge of the firm, thus internal knowledge-oriented approach enables the firm to distinguish itself from its rival firms.	Thornhill (2006)	SMEs	Manufacturing
		Zhou and Li (2012)	Large firms and SMEs	High technology
External knowledge-oriented	External knowledge helps firms to develop a broader knowledge base so that a firm can keep abreast of emerging technologies;	Fey and Birkinshaw (2005)	Large firms	Chemical, electric, electricity, food, machinery, pulp and paper, Transport equipment
		Kang and Kang (2014)	Large firms and SMEs	Service
	External knowledge-oriented approach reduces costs to develop internal knowledge base and increase the firms' speed of implementing its newly acquired knowledge.	Vega-Jurado <i>et al.</i> (2009)	Large firms and SMEs	Manufacturing
		Zhao <i>et al.</i> (2005)	Large firms and SMEs	Manufacturing
Open innovation (both internal knowledge- and external knowledge-oriented)	Open innovation (pursuing internal knowledge- and external knowledge-oriented approaches together) allows a firm to draw on a wide range of knowledge sources and adaptive response.	Cassiman and Veugelers (2006)	Large firms and SMEs	Manufacturing
		Chen <i>et al.</i> (2016)	Large firms and SMEs	Car manufacturing, electrical and communication equipment, food, materials and chemical, metallic mineral products, paper and furniture, pharmaceuticals, textile and clothing
	Open innovation enables a firm to exploit existing knowledge and explore a broad set of external knowledge, leading to improved performance.	Hoang and Rothaermel (2010)	Large firms	Pharmaceuticals
		Prabhu <i>et al.</i> (2005)	Large firms and SMEs	Pharmaceuticals